

Usability Report

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1 Introduction

Evaluation of qualitative aspects of the Physio Companion app was done through a usability study. The project timeline and resources limit the participant pool size, and as a result, this study does not provide any quantitative insights. This document outlines the testing methodology and concludes with the resulting findings.

The goals of the study are:

- Assess the ease of use of the app across a diverse demographic
- User understanding of the terminology within the app
- User perception of the flow and navigation within the app
- Identification of any friction points
- Feedback on additional and beneficial features within the app

2 Techniques

The techniques used to conduct these tests were a mixture of both in-person and remote moderated testing. Data was collected via a questionnaire or an interview. The choice was up to the discretion of the investigator, but both formats covered the same core topics. The participants were first asked a series of questions inquiring some background information. They were then shown how to install the app on their smartphone. After successful installation of the program, the facilitator prompted the participant on the app's use. An additional investigator was playing the role of an observer recording the user's behaviour. At the end of the study, participants were asked for their final impressions. The study was voluntary and lasted approximately 30-40 minutes.

3 Methodology

In order to gain a deeper understanding of the user experience of the application, a series of questionnaires were constructed for each stage of testing ([pre-session](#), [session](#), and [post-session](#)). In addition, the 'session' questionnaire was further subdivided in order to examine the different main actions of the application. This breakdown is crucial to understanding the different facets of the application. Obtaining feedback promptly after completing each section will allow immediate and reliable reporting of the participants' impressions. The questionnaires were constructed as a guide and investigators were free to use either an interview-style approach or use the explicit questionnaire. If the former was used, it was essential that the topics mentioned in the questionnaire were addressed.

4 Participant profiles

The participant pool comprises of individuals of varying ages, genders and occupations, which ensures coverage of a broad target user base. The coverage details are outlined below:

- Name
- Age: Participants of various ages were chosen, with the sole restriction being that they had to be over 18.
- Gender: In an ideal case, an equal and all-encompassing balance of genders would be sought out. This would include 2 female or male identifying individuals and one non-binary individual.
- Person with disability: Another criteria that ensures diversity is enlisting participants with disabilities. This will aid in identifying any accessibility barriers that may go unnoticed. In an ideal case, at least 1 or 2 participants would be those who have a disability.
- Occupation: This criteria was selected to give insight into lifestyle factors. For the purposes of this study, any bias towards conducting it for the same occupation would skew the results.
- Level of digital literacy: This measure pertains to smartphone usage. It is divided into low, medium, or high expertise indicating how comfortable the user is operating the device.
- Familiarity with injury to the knee: This relates to how familiar the participant is with a knee injury, whether they are familiar with it through their occupation (e.g. orthopaedic surgeon, physiotherapist) or whether they have had a knee injury in the past.

Participant 1

- Age: Late 20s
- Gender: Male
- Person with disability: No
- Occupation: Physiotherapist
- Level of digital literacy: High
- Familiarity with injury to the knee: Very familiar

Participant 2

- Age: Late 20s
- Gender: Male
- Person with disability: No
- Occupation: Physiotherapist
- Level of digital literacy: High
- Familiarity with injury to the knee: Very familiar

Participant 3

- Age: early 60s
- Gender: Male
- Person with disability: No
- Occupation: Technician
- Level of digital literacy: High
- Familiarity with injury to the knee: Familiar

Participant 4

- Age: 24
- Gender: Female
- Person with disability: No
- Occupation: Student
- Level of digital literacy: High
- Familiarity with injury to the knee: Not familiar

Participant 5

- Name
- Age
- Gender
- Person with disability
- Occupation

- Level of digital literacy
- Familiarity with injury to the knee

Due to project timeline constraints, the participant pool was limited in size, and posed challenges in recruiting individuals who satisfied the criteria mentioned above. Furthermore, it was not possible to recruit a participant with a knee injury. However, there were some that had undergone physiotherapy treatment for other health reasons.

Two of the participants were physiotherapists. They were recruited to assess the PT flow of the application and provide their perspective. The remaining participants varied in age and fulfilled different combinations of the criteria above. Some of the participants were familiar with what 'physiotherapy' entails even though none of them have had a knee surgery before. These were the two physiotherapists that were interviewed and the participant 3, who had to perform single leg flexion and extension as part of their home exercise program.

5 Testing

All questionnaires' accompanying logic is explained [here](#).

5.1 Pre-session questionnaire

See Appendix [B](#).

5.2 Session questionnaire

See Appendix [C](#)

5.3 Post-session questionnaire

See Appendix [D](#).

6 Results

Physiotherapists

These points emerged as common feedback for both of the physiotherapist participants (See section Appendix [A](#)).

- Subjective experience performing exercise: participants emphasized the inclusion of an RPE (Rate of Perceived Exertion) from the patient as an additional measure.
- Additional exercise information: They also suggested incorporated the number of sets (in addition to reps) performed in each exercise session on the Feedback tab.

Patients

The data obtained from the 'patient' participants highlighted the observations listed below.

Common to participants:

- Desire for audio/visual cues: Participants explicitly stated that they had some trouble looking at the screen while exercising. They emphasized that audio cues would be a beneficial addition to the exercise recording function. In addition to that, they suggested enabling customizable font settings. These would benefit if there was a change in the ambient environment that prevented clear viewing of the textual cues.
- Intuitive navigation: Participants agreed that the app was easy to navigate and required minimal effort to execute a desired action.
- Engagement and motivation: Participants found the app engaging and noted that it may be a good motivator for completing home exercises.

Unique feedback:

- Pose detection: One participant noted that the app sometimes had trouble detecting movement that resulted in delays.
- Feedback speed: Delays in pose or form detection did not impact the speed with which on screen feedback was presented.
- One participant stated that a simplified summary (in plain language) would be helpful. A quick user-friendly overall progress summary in plain language would be preferred to viewing raw data charts.

Feedback speed vs pose detection is insightful because ?

7 Discussion

Usability testing proved to be an invaluable asset in evaluating the application. The results were enlightening and were immensely beneficial to the development team in refining the final product.

For the physiotherapist participants, the study was conducted in a conversational interview format, because it proved to be more natural than a questionnaire. The physiotherapists tested the PT flow of the application and their input was particularly insightful. They noted that the patient's subjective experience was an indispensable measurement, which is extremely valuable when determining exercise modifications. They provided examples of how this may indicate how difficult or easy the exercise is for the patient and whether their performance is impacted by other limitations (eg. pain, frequency of exercise performance). Including the number of sets in addition to the number of reps is useful for the same reason. Both participants found the navigation and its use within the application to be straightforward. Their positive response to the application suggests that it has strong potential and can be advantageous to patient recovery.

Reception of the app by the participants playing the role of 'patients' was well-received. They remarked on the intuitive navigation, clear layout and enhanced engagement made the app's use seamless and enjoyable. However, important accessibility concerns were identified. The feedback provided on the device's screen was difficult to read at the distance that the app required for the participant's proper form detection. Additionally, the absence of audio necessitated frequent glances to the device's screen, which disrupted focus and optimal performance of the exercise. These findings suggest that, while the interface within the app itself is easy to follow, the real-time exercise guidance layer would benefit from further development.

Of note, an interesting dichotomy emerged within one aspect of the findings. One participant reported issues with the overall pose detection functionality, noting that the application experienced difficulty detecting movement and body parts, which may have introduced occasional delays. Conversely, the other participant raised no concerns regarding pose detection, found the feature satisfactory, and specifically remarked on the speed of feedback, describing it as instantaneous. A range of factors may account for this distinction, including variations in ambient environmental conditions during exercise performance, differences in internet connectivity stability, or potential packet transmission irregularities since image frames were being sent to the backend to be processed.

8 Conclusion

A PT feedback

A.1 Usability Test Feedback: Physiotherapist 1

- **On a scale of 1 to 10, how easy was it for you to log in?**
Response: "Pretty good. Like, no complaints. Yeah, pretty straightforward."
- **Tell me what you see and what you think is going on in this page [the homepage]?**
Response: Recognized it as a welcome page. Correctly identified that linked patients are listed on the left and the patient activity history shows recent exercises.
- **From this page, can you tell me what an "invite code" is?**
Response: "No." The moderator then explained it is for linking patients to the physiotherapist's organization.
- **Was there any sort of confusion when you initially landed on the homepage, and what would make it easier to understand?**
Response: The "patient activity history" on the main dashboard feels slightly repetitive since that information is also available in the feedback section. He would prefer to swap out the activity history on the main dashboard with a direct list of the specific exercises assigned to that patient, as that is what physiotherapists look at the most.

- **Is there a specific organizational structure that would help you to better address your patients (e.g., categorizing by most recent patient, submission, or exercise)?**

Response: The current format looks good, but he recommended removing the "Recommended" versus "Optional" categorization. Exercises should be treated like prescribed medication—patients just have to do them. It is best to keep it simple and just list the required exercises.

- **Do you feel like you have enough information in the exercise logs to make decisions or give comments about a patient's progress?**

Response: Requested the addition of "Sets" and "Reps" (e.g., 10 reps for 2 sets), as well as "Holds" (hold duration for a stretch). This provides a much more realistic picture of what the patient is doing.

- **Is there anything particular that you found was hard to do when modifying the instructions?**

Response: No, but having a physically separate input box for assigning "Reps" and "Sets" (rather than just typing it into a general instruction text box) would be perfect.

- **If we added a second exercise, would you prefer a list of every available exercise or an "add exercise" dropdown?**

Response: An "add exercise" dropdown, if possible.

- **How useful would you say are the charts and graphs for tracking a patient's progress over time?**

Response: Mildly helpful. He noted that he currently goes mostly by what the patient tells him. However, he strongly recommended keeping it in the app because it's a great extra feature that helps the app stand out from competitors.

- **Was there any data that you expected to see here that is missing?**

Response: It would be highly helpful to track patient effort levels, specifically RPE (Rate of Perceived Exertion), and pain levels on a scale of 0 to 10. This is especially crucial for post-op patients to ensure they aren't pushing too hard and causing excruciating pain.

- **Do you feel that it's better to remove all patient history once a user is deleted, or have a log of previous patients?**

Response: By law, healthcare professionals must keep medical records for about 8 years. Physiotherapists generally never delete patients.

- **Follow-up Question: Would it be safe to say we can have an "offload" or "inactive" feature instead of "delete"?**

Response: Yes. Labeling it as an "inactive account" or "discharged" is a good idea.

- **On a scale of 1 to 10, how easy was the app to navigate without any instructions?**

Response: A solid 8.5.

- **What was the most confusing part of the app?**

Response: The feedback section wasn't super obvious at first glance. He suggested adding a quick blurb or context at the top of the screen, such as: "Click on an exercise to leave your patients a comment."

- **If there was a feature you would add or change to make this more useful in your practice, what would that be?**

Response: An easy way to add and remove exercises. Physiotherapists spend 99% of their time prescribing and modifying exercises, and it is the biggest hassle at the end of the day.

- **For tracking, is it required for the app's camera to automatically measure sets/holds, or is tracking total completed reps alongside the requested reps/sets enough?**

Response: Displaying the requested reps/sets and only having the app count the completed reps is perfectly fine. The app does not need to automatically measure hold lengths or sets.

A.2 Usability Test Feedback: Physiotherapist 2

- **On a scale of 1 to 10, how easy was it for you to log in?**

Response: Pretty straightforward.

- **Tell me what you see and what you think is going on in this page [the homepage]?**

Response: Explained the main dashboard correctly. Noted that the patients listed are his patients, and the history section is to see whether they are doing their assigned physio. He assumed that the invite code was to be sent to a new patient to add them to this application.

- **From this page, can you tell me what an "invite code" is?**

Response: Assumed correctly—the invite code is to be given to a new patient, who can then enter it on their side to join the physiotherapist's planning.

- **Was there any sort of confusion when you initially landed on the homepage, and what would make it easier to understand?**

Response: Overall, it was pretty straightforward. He did note, however, that clicking a patient to view their history was not immediately obvious—he did not do that himself without prompting.

- **Is there a specific organizational structure that would help you to better address your patients (e.g., categorizing by most recent patient, submission, or exercise)?**

Response: Looks pretty good. I think adding more information would be helpful (e.g. frequency per day, sets)

- **Is there anything particular that you found was hard to do when modifying the instructions?**

Response: No, it was pretty self-explanatory. He suggested that adding pictures or a visual component would make it easier on their end. Additionally, he reiterated that including sets, reps, and holds would be beneficial.

- **If we added a second exercise, would you prefer a list of every available exercise or an "add exercise" dropdown?**

Response: "I think having some sort of search bar would be easier to navigate through the exercises."

- **How useful would you say are the charts and graphs for tracking a patient's progress over time?**

Response: He emphasized the need to track not just sets, but also which specific exercise was performed. For example, if the rep count is 30, it is unclear whether that was a 3x10 or a 2x15, or what exercise it corresponds to. He also suggested including RPE (Rate of Perceived Exertion).

He placed a big emphasis on distinguishing between multiple exercises (e.g., upper body vs. lower body), noting that an overall rep count becomes less meaningful when exercises are mixed.

- **Was there any data that you expected to see here that is missing?**

Response: He highlighted the value of granular data—for example, if a patient did 3x10 and the first two sets went well but on the third set, the sixth rep was difficult, flagging that specific moment would be more important to a physio. This level of detail allows them to tailor the exercise regimen more effectively (e.g., identifying if the load is too much or too little).

- **Do you feel like you have enough information in the exercise logs to make decisions or give comments about a patient's progress?**

Response: "Not really." While objective numbers are nice, having subjective comments from the user would be much more helpful from a physio's point of view. Again, he mentioned that tracking RPE and how the patient felt during exercises would be valuable.

- **Do you feel that it's better to remove all patient history once a user is deleted, or have a log of previous patients?**

Response: Deleting is fine, so long as there is some sort of export function. He expressed his frustration with how long charting currently takes. It would be ideal to export

patient notes (e.g., as a PDF) directly into their record-keeping application. This way, the practice could keep records without needing to retain the patient’s account indefinitely.

- **On a scale of 1 to 10, how easy was the app to navigate without any instructions?**

Response: Overall, it was pretty straightforward, and the UI was easy to use. However, he noted that within the feedback section, adding a visual indicator (e.g., a ”>” icon) would help show that tiles are clickable—he had to be told they were interactive.

- **What was the most confusing part of the app?**

Response: Mentioned above (clickable tiles were not immediately obvious).

- **If there was a feature you would add or change to make this more useful in your practice, what would that be?**

Response: Mentioned above (export functionality, visual indicators for clickable elements, and more granular exercise tracking).

B Pre-session Questionnaire

This questionnaire is designed for the patient participants of the study to obtain to obtain some preliminary information about the their experience with a knee injury.

It will be administered in a structured manner, where participants will be presented the questions in the order they were designed. This will ensure consistency and that the essential amount of background detail is relayed to the investigators.

B.1 Patient Experience Information

1. What is your name?
2. How old are you?
3. What gender do you identify with?
4. What is your occupation?
5. Have you ever had a knee injury? **Yes/No/NA**
 - (a) If **YES**, was it in the past 5 years?
 - (b) If **YES**, have you had physiotherapy after your knee surgery?
 - (c) If **NO**, have you had another kind of orthopaedic surgery?
 - (d) If **NA**, do you personally know anyone who has experienced a knee injury?
YES/NO

6. How long did your post-operative physiotherapy last?
7. Were you prescribed knee extensions as an exercise to do at home?
8. In your opinion, did you encounter any trouble performing knee extensions? **YES/NO**

B.2 DEVICE LITERACY

9. What do you use your mobile device for?
10. How comfortable are you using the camera function?

Very comfortable	Somewhat comfortable	Comfortable
Not comfortable		
11. Do you know what a mobile phone tripod is? (selfie stick)
12. How comfortable are you using the tripod?

Very comfortable	Somewhat comfortable	Comfortable
Not comfortable		

B.3 PRIVACY

This section outlines some private aspects involved in conducting a usability test. These statements should be confirmed with participants to ensure that a well-informed decision to participate has been made.

1. Participant identification information will be stored on a database.
2. Information extracted from the recording used in the application will also be stored on a database.
3. The database used in the application adheres to data privacy standards and regulations listed [here](#).
4. All participant data will be manually deleted by April 8th, 2026.
5. Information access is restricted to any external parties (except the application developers for the duration mentioned above).
6. All participant data will be manually deleted by April 8th, 2026.

C Session Questionnaire

This document contains questions that participants will be asked by the investigator. These questions provide a roadmap of the application functionalities and investigators are free to pursue their own line of questioning regarding these functionalities. Throughout the application's testing, the investigator will time the duration spent on each section. Details such as timing justification, survey duration and participant information can be found in the Usability Report.

C.1 Creating an account

After participant verification, the application will be run on the investigator's device. Note that the participant are familiar with operating the device used and will be asked to launch the application and proceed through each stage below.

Please click on 'create account' to be redirected to the account creation functionality. An invite code/PT license number should be provided to the participant. The duration to accomplish this task should be recorded.

*If the participant selects the **NO** option for any **YES/NO** questions, please inquire further.*

C.1.1 UI

1. Are the different elements on the screen logically positioned? **YES/NO**
2. Is the text font size sufficient for reading?
3. Can you clearly distinguish between the different colours on the screen?
4. Are the buttons appropriately sized?

C.1.2 Functionality

1. Rate the difficulty level for creating an account: low, medium, high
Very easy Easy Somewhat difficult Difficult
2. Do you have any comments/improvements for account creation?

Once the account has been created, the next step will be navigation to the home screen. Initially, the home screen will be empty. It will be populated once an exercise has been recorded.

C.2 Homescreen

This is the page that is navigated to after the user has successfully signed in.

C.2.1 UI

1. Are the visual elements arranged logically? **YES/NO**
2. Are those elements clearly defined? **YES/NO**
3. Can you read the text on the screen without difficulty? **YES/NO**
4. Are the functions (on the bottom) of each button clearly defined? **YES/NO**
5. Any comments/improvements for this?

C.3 Recording an exercise

Participant will click on the button at the bottom of the homescreen which will lead them to the 'Record' page

C.3.1 UI

1. Are the visual elements arranged logically? **YES/NO**
2. Are those elements clearly defined? **YES/NO**
3. Can you read the text on the screen without difficulty? **YES/NO**
4. Are the functions of each button clearly defined? **YES/NO**
5. Do the icons navigate to their correct page? **YES/NO**
6. Any comments/improvements for this?

C.3.2 Functionality

1. Is the app displaying detectable area (of the form) appropriately? **YES/NO**
The participant will proceed to perform the exercise once they are in position
2. Is the app effectively detecting the relevant body part? **YES/NO**
3. Is the app detecting form discrepancies (angle too small etc) to a high level of accuracy?
YES/NO
4. Is the app displaying position modification instructions clearly and effectively? **YES/NO**
5. Is each rep being detected correctly/accurately? **YES/NO**
6. In your opinion, how accurate do you think the detection is? **YES/NO**
7. Any comments/improvements for this?

C.4 Exercise page

The participant will be asked to navigate to the 'Exercises' page.

C.4.1 UI

1. Are the visual elements arranged logically? **YES/NO**
2. Are those elements clearly defined? **YES/NO**
3. Can you read the text on the screen without difficulty? **YES/NO**
4. Do the icons navigate to their correct page? **YES/NO**
5. Any comments/improvements for this?

C.4.2 Functionality

1. Are the instructions clear and easy to follow without being overwhelming? **YES/NO**
2. Is the example video showing how to perform the exercise helpful? **YES/NO**
3. Any comments/improvements for this?

C.5 Progress

C.5.1 UI

1. Are the visual elements arranged logically? **YES/NO**
2. Can you read the text on the screen without difficulty? **YES/NO**
3. Any additional information that could be included?

D Post-session Questionnaire

These are more open-ended questions to be asked at the end of the session. Like the questionnaires used previously, the investigators are free to follow their own line of questioning provided they include the details below.

D.1 Patients

1. Would this be a tool that you use in the recovery process (from a knee injury in this case)?
2. Can you think of any improvements? (if they say no, probe further. eg. what are their thoughts on UI, pose detection etc)

3. What did you like about the application?

D.2 PTs

1. Would you use an application like this in your practice?
2. Can you think of any improvements? (if they say no, probe further. eg. what are their thoughts on UI, pose detection etc)
3. What did you like about the application?

Please remember to thank the participants for their time and invite them to the EXPO on April 7th.